

Penetration Depth of a Laser into Materials

Source: <https://www.globalsino.com/EM/page881.html>

The penetration depth of a laser into materials is primarily governed by the absorption coefficient of the material at the laser's wavelength. For many materials, this absorption is wavelength-dependent, and the penetration depth d is related to the absorption coefficient α as,

$$d = \frac{1}{\alpha} \quad (881a)$$

where,

- d is the penetration depth (the depth at which the intensity of the laser beam drops to 1/e or about 37% of its original intensity).
- α is the absorption coefficient of the material at a given wavelength (usually in cm^{-1}).

The absorption coefficient depends on the material's properties (such as bandgap, phonon interactions) and the wavelength of the laser. For some common laser wavelengths (e.g., 1064 nm for Nd lasers or 532 nm for green lasers), materials will have different absorption coefficients. If we know the material's refractive index n and extinction coefficient k at the laser wavelength, we can calculate the absorption coefficient using the following relation,

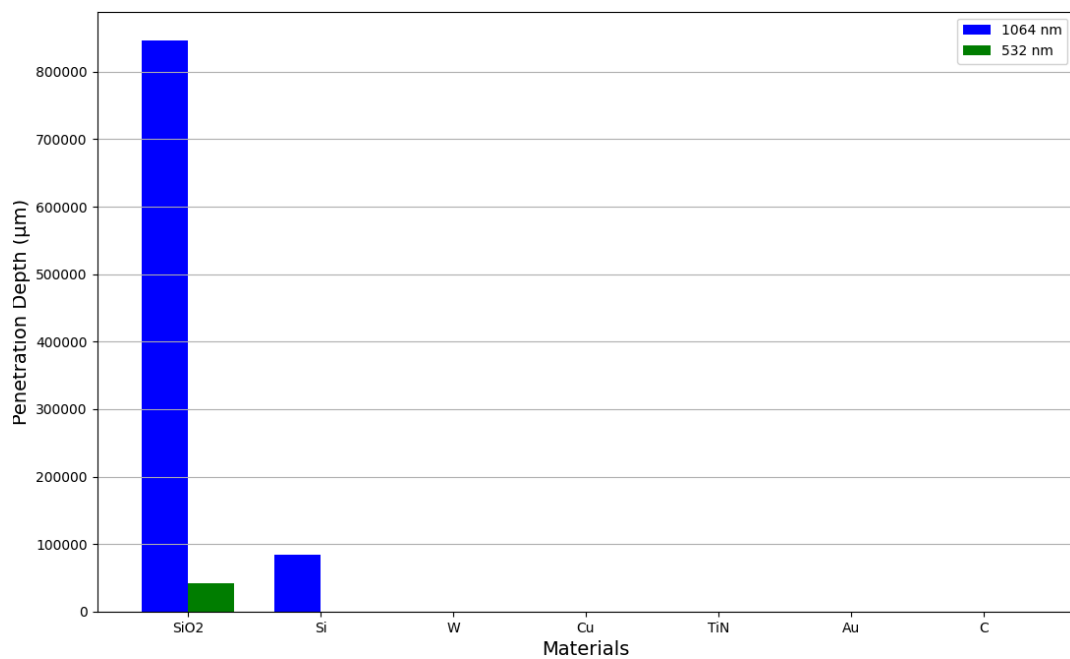
$$\alpha = \frac{4\pi k}{\lambda} \quad (881b)$$

Where,

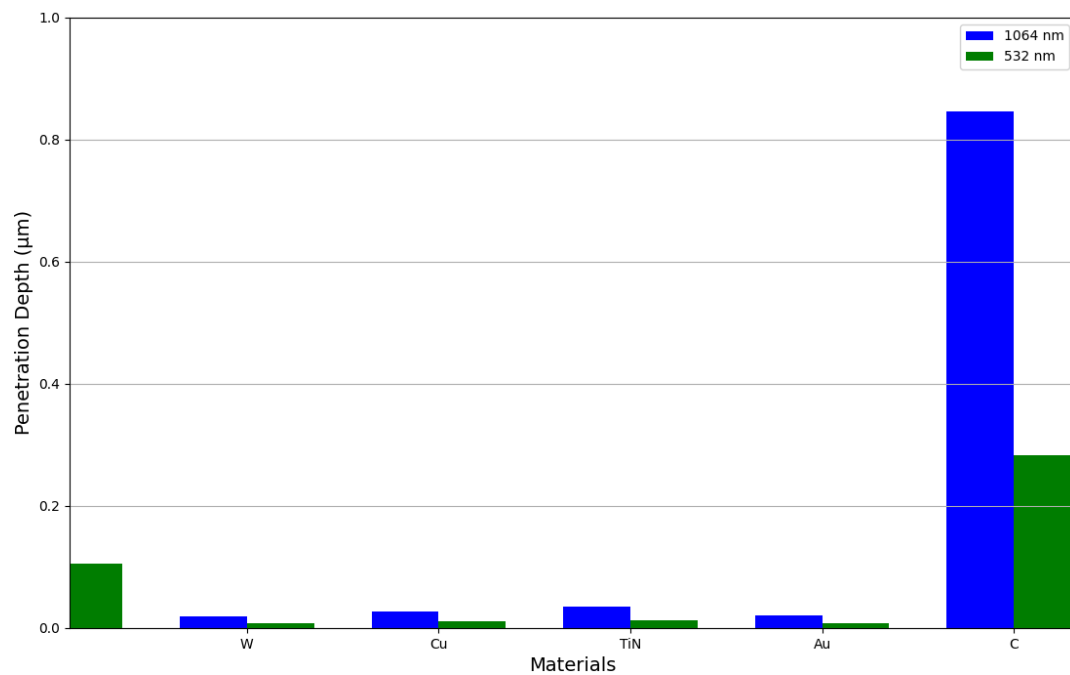
- k is the extinction coefficient (imaginary part of the complex refractive index).
- λ is the wavelength of the laser in the same units as α .

At 1064 nm (infrared), silicon (Si) is transparent at this wavelength, so the penetration depth is large (several millimeters); at 532 nm (green), silicon absorbs strongly, and the penetration depth is in the range of microns ($\sim 3 \mu\text{m}$). For silicon dioxide (SiO_2), at 1064 nm: SiO_2 is transparent, with penetration depths in the millimeter range. For copper (Cu), at 1064 nm, the absorption coefficient is high due to strong absorption, and the penetration depth is very small (on the order of nanometers). For gold (Au), at 532 nm, Au has a very high absorption coefficient, and the penetration depth is on the order of tens of nanometers. For tungsten (W), at 1064 nm, tungsten absorbs efficiently, and the penetration depth is very small (nanometer range).

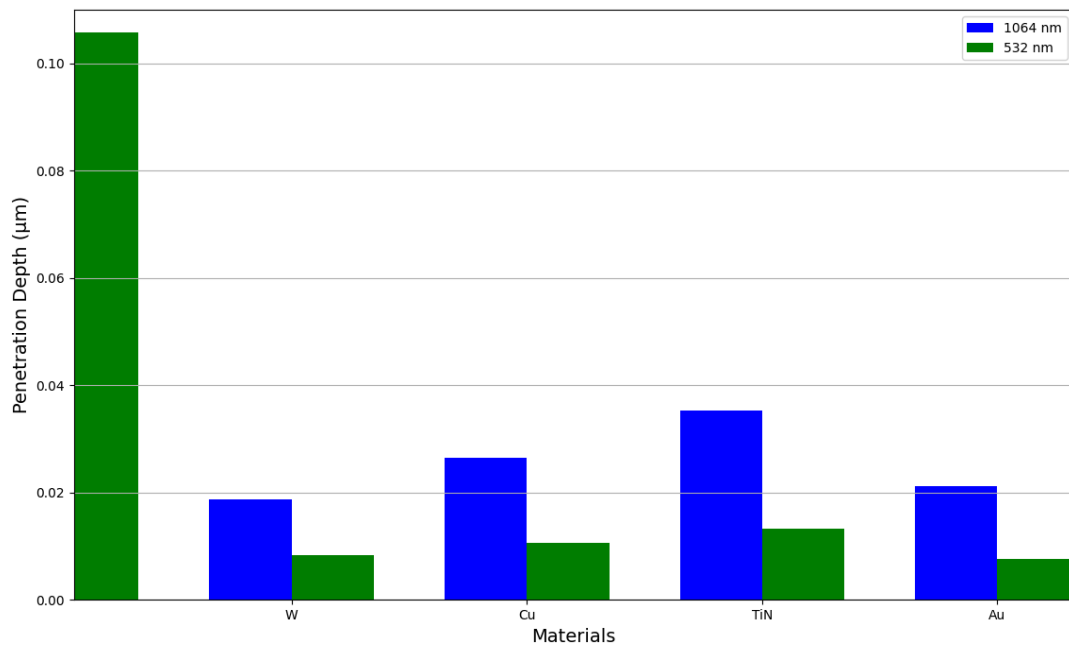
Figure 881a shows the laser penetration depths at the laser wavelengths of 1064 nm and 532 nm, respectively.



(1/3) 전체 범위: SiO₂, Si, W, Cu, TiN, Au, C



(2/3) 세로축 0-1 μm 확대 (왼쪽 끝의 잘린 녹색 막대는 Si, 532 nm)



(3/3) 세로축 0–0.1 μm 확대: W, Cu, TiN, Au

Figure 881a. Laser penetration depths at the laser wavelengths of 1064 nm and 532 nm, respectively.

※ 패넬별 설명(1/3~3/3)은 원문에 없는 편집 주석. 원문은 세 그래프를 캡션 하나로 묶어 제시함. 수식 [881a], [881b]는 원문에서 이미지였던 것을 같은 내용으로 다시 조판함.